

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q21-1812

Spring Gap Tap 138 kV
August 21, 2022

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q21-1812. The Interconnection Customer proposes a 213.0 MW solar plus storage facility interconnecting at the Spring Gap Tap 138 kV substation in the Virginia area. The request was received on July 10, 2021. The study evaluated system intact and single contingency conditions using the regional planning case, including 607 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 21 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$38,831,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

The VAN TYLE SUBSTATION – RIVERTON 230 138 kV is overloaded to 129.0% of its 344 MVA emergency rating for the loss of the MUSTANG – ELECTRIC AVENUE FACILITY 138 kV. The project contributes 9.0% DFAX (19.1 MW) to this overload.

For the loss of W STAUNTON – WOODS P 138 kV, the STONEWALL ENERGY CENTER – BLAINE 138 kV loads to 125.4% of its 329 MVA rating. Project DFAX: 5.1%.

The SPRING GAP TAP SUBSTATION – MUSTANG 138 kV is overloaded to 123.8% of its 271 MVA emergency rating for the loss of the OAK GREEN – SPRING GAP TAP 138 kV L. The project contributes 59.5% DFAX (126.6 MW) to this overload.

CHAFFEE GAS RECOVERY – BLAINE 138 kV: 121.7% loading (371 MVA) following the outage of the WOODS DP – W STAUNTON 138 kV; DFAX = 5.1%.

The LEWISBURG – SOUTH JORDAN 138 kV is overloaded to 119.6% of its 343 MVA emergency rating for the loss of the MIDDLETOWN JCT – MOUNT PLEASANT 138 kV line. The project contributes 6.8% DFAX (14.5 MW) to this overload.

The OAK GREEN – SPRING GAP TAP 138 kV line is overloaded to 117.3% of its 288 MVA emergency rating for the loss of the SPRING GAP TAP – MUSTANG 138 kV. The project contributes 59.1% DFAX (126.0 MW) to this overload.

The S JORDAN – FIREBANTZ 138 kV L is overloaded to 114.0% of its 203 MVA emergency rating for the loss of the MOUNT PLEASANT – FIREBANTZ 138 kV. The project contributes 5.1% DFAX (10.8 MW) to this overload.

MUSTANG – ELECTRIC AVENUE FACILITY 138 kV line: 112.3% loading (337 MVA) following the outage of the VAN TYLE – RIVERTON 230 SUBSTATION 138 kV line; DFAX = 15.4%.

CHAFFEE GAS RECOVERY – SOUTH JORDAN 138 kV: 111.5% loading (415 MVA) following the outage of the WHEELER – PASCO CNTY SOLID WASTE RESOURCE RECOVERY 138 kV Line; DFAX = 5.7%.

VAN TYLE – COYOTE CRK SUBSTATION 138 kV Line: 109.6% loading (313 MVA) following the outage of the BIRCHWOOD SUBSTATION – LEAVITT 345 kV line; DFAX = 9.5%.

For the loss of NAUVOO – BIRCHWOOD 345 kV Line, the NESTUCA – POLK CITY 345 kV loads to 107.5% of its 1,458 MVA rating. Project DFAX: 5.7%.

For the loss of MOUNT PLEASANT – MIDDLETOWN JCT 138 kV, the INDECK WEST ENFIELD ENERGY CENTER – LEWISURG 138 kV loads to 105.4% of its 184 MVA rating. Project DFAX: 8.6%.

The FIREBANTZ SUBSTATION – WHEELER 138 kV is overloaded to 104.7% of its 491 MVA emergency rating for the loss of the S JORDAN – CHAFFEE GAS RECOVERY 138 kV line. The project contributes 9.6% DFAX (20.5 MW) to this overload.

TOA VACA SUBSTATION – THOMASTON 138 kV: 104.3% loading (343 MVA) following the outage of the BLAINE – CHAFFEE GAS RECOVERY 138 kV Line; DFAX = 5.7%.

The NAUVOO – POSSUM POINT 345 kV is overloaded to 103.6% of its 1,479 MVA emergency rating for the loss of the NAUVOO – BIRCHWOOD 345 kV line. The project contributes 5.8% DFAX (12.3 MW) to this overload.

For the loss of THOMASTON – CARRABELLE 345 kV Circuit 2, the CARRABELLE – THOMASTN 345 kV Cir. 1 loads to 103.2% of its 1,435 MVA rating. Project DFAX: 8.5%.

The THOMASTON – CARRABELLE 345 kV is overloaded to 103.2% of its 1,435 MVA emergency rating for the loss of the THOMASTON – CARRABELLE 345 kV Ckt 1. The project contributes 8.5% DFAX (18.2 MW) to this overload.

MOUNT PLEASANT – FIREBANTZ SUBSTATION 138 kV L: 103.0% loading (412 MVA) following the outage of the S JORDAN – CHAFFEE GAS RECOVERY 138 kV; DFAX = 6.7%.

ROBERTS SW STA – NAUVOO 345 kV: 102.2% loading (2,021 MVA) following the outage of the CITY W DIESEL PLANT – NAUVOO SUBSTATION 345 kV; DFAX = 7.1%.

CONTRA COSTA – LIBBY 138 kV: 100.6% loading (655 MVA) following the outage of the NAUVOO – BIRCHWOOD 345 kV Line; DFAX = 6.0%.

For the loss of LOUISA GENERATION FACILITY – N ANNA 345 kV, the CARRABELLE – LOUISA GENERATION FACILITY 345 kV loads to 100.5% of its 2,525 MVA rating. Project DFAX: 12.4%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
VAN TYLE – RIVERTON 230 138 kV	Rebuild line with higher-capacity conductor	\$5,324,000	43% (\$2,310,000)
STONEWALL ENERGY CTR – BLAINE 138 kV	Rebuild line with higher-capacity conductor	\$1,567,000	9% (\$137,000)
MUSTANG – SPRING GAP TAP 138 kV L	Rebuild line with higher-capacity conductor	\$16,557,000	59% (\$9,804,000)
CHAFFEE GAS RECOVERY – BLAINE 138 kV	Rebuild line with higher-capacity conductor	\$2,210,000	9% (\$193,000)
LEWISBURG – SOUTH JORDAN 138 kV	Rebuild line with higher-capacity conductor	\$7,317,000	28% (\$2,037,000)
OAK GREEN SUBSTATION – SPRING GAP TAP 138 kV	Rebuild line with higher-capacity conductor	\$7,466,000	59% (\$4,401,000)
S JORDAN – FIREBANTZ 138 kV L	Rebuild line with higher-capacity conductor	\$13,663,000	22% (\$3,080,000)
MUSTANG – ELECTRIC AVENUE FACILITY 138 kV line	Rebuild line with higher-capacity conductor	\$8,812,000	49% (\$4,325,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
CHAFFEE GAS RECOVERY – S JORDAN 138 kV	Rebuild line with higher-capacity conductor	\$28,100,000	12% (\$3,471,000)
COYOTE CRK – VAN TYLE 138 kV	Reconductor line	\$5,201,000	42% (\$2,160,000)
NESTUCA – POLK CITY 345 kV	Reconductor line	\$7,658,000	2% (\$159,000)
INDECK WEST ENFIELD ENERGY CTR – LEWISBURG 138 kV	Reconductor line	\$5,190,000	54% (\$2,785,000)
FIREBANTZ – WHEELER 138 kV L	Reconductor line	\$2,101,000	11% (\$227,000)
THOMASTON – TOA VACA 138 kV	Reconductor line	\$805,000	6% (\$49,000)
NAUVOO SUBSTATION – POSSUM PT 345 kV line	Reconductor line	\$12,486,000	2% (\$270,000)
CARRABELLE – THOMASTN 345 kV Cir. 1	Reconductor line	\$17,993,000	5% (\$859,000)
THOMASTON – CARRABELLE 345 kV	Reconductor line	\$26,678,000	5% (\$1,274,000)
MOUNT PLEASANT – FIREBANTZ 138 kV	Reconductor line	\$2,128,000	10% (\$216,000)
NAUVOO – ROBERTS SWSTA SUBSTATION 345 kV line	Reconductor line	\$20,857,000	1% (\$255,000)
CONTRA COSTA – LIBBY 138 kV	Reconductor line	\$2,121,000	12% (\$249,000)
CARRABELLE – LOUISA GENERATION FACILITY 345 kV line	Reconductor line	\$26,020,000	2% (\$570,000)
	Total allocated to project		\$38,831,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.